How humans solve indoor orienteering courses

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Abstract

Algorithmic analysis of indoor orienteering courses models the venue as a graph and finds routes by exhaustive search. A competitor reading a map mid-race cannot explore the whole graph, holds only a handful of items in working memory, and must commit in seconds — yet solves within-floor navigation almost instantly, at a glance. This article develops a cognitive account of human route finding: within-floor regions are solved perceptually as single units of reasoning, so the human's effective search space is the condensed *cluster graph* of floor regions connected by labeled stairs. On that graph, humans plan bidirectionally from both controls, prefer near and goal-directed stairs, and commit to the first connection found rather than the best one. We analyze the challenges this strategy faces — one-way passages, discontinuous map segments, working-memory overflow — and the training methods that address them, and describe a deterministic simulation that enables new difficulty measures and empirical tests.

1 Introduction

A companion article introduces a graph model of indoor orienteering venues: nodes at decision-relevant positions, edges for corridors and stairs, one-way constraints, and floor membership resolved from level polygons; the identity of a route is its sequence of labeled stair crossings [1]. On this model, algorithms find the shortest route by Dijkstra search and can enumerate *every* distinct route between two controls, shortest first.

A human competitor holding the map does none of this. They cannot enumerate; they often cannot even verify optimality of the route they choose. Still, experienced indoor orienteers solve legs quickly and mostly well. Two observations frame this article:

1. **Within a floor, route finding is perceptual, not deliberative.** Given a start and a target on the same map segment, a quick glance is usually sufficient: the runner *sees* a path through the corridors the way one sees the way out of a drawn maze, without item-by-item reasoning.

2. **Across floors, route finding is deliberative and memory-bound.** Choosing which staircases to take, in which order, is slow, error-prone, and limited by how many intermediate steps the runner can hold in mind while also running.

The venue-graph model explains why the boundary falls exactly there, what the human's effective search problem is, and which strategies make it manageable. Understanding this has three uses: it predicts where courses are hard (a difficulty model should count *mental* work, not graph edges); it suggests concrete training methods; and it can be simulated, giving analysis tools a "human" route to compare against the optimal one.

2 Background

Research on wayfinding in multi-level buildings shows that vertical transitions dominate difficulty and that people adopt identifiable strategies. Hölscher et al. [2], studying a complex multi-level building, distinguish a *central-point* strategy (route via well-known parts), a *direction* strategy (head toward the horizontal position of the goal first), and a *floor* strategy (get to the goal's floor first); experienced participants favored the floor strategy. Wiener et al. [3] classify wayfinding tasks by the knowledge available; map-based indoor orienteering is unusual in giving complete survey knowledge (the map) under severe time pressure.

Cognitive constraints are well established: working memory holds only about four chunks [5] (the classical estimate was seven [4]), and expert advantage in domains like chess comes largely from *chunking* — perceiving familiar configurations as single units [6]. Decision makers under time pressure *satisfice*: they take the first option that clears an acceptability threshold rather than optimizing [7]. In orienteering specifically, expert cognition centers on anticipatory map reading and strategies that keep attention free for execution [8].

Indoor orienteering maps add a further, unusual burden: each floor is printed as a separate map segment, so a single staircase appears *twice* — once on each floor's segment — linked only by its label. Following a route across floors therefore requires repeated visual search across segments, a task with no outdoor counterpart.

3 The human search problem on the venue graph

3.1 Floor regions are solved perceptually

The algorithmic framework condenses every region reachable through ordinary two-way corridors into a single *cluster*, because corridors never change route identity [1]. The cognitive claim of this article is that the same condensation happens perceptually in the human solver:

> A floor region visible on one map segment is, for a competent map reader, a solved subproblem — a single chunk. Route finding within it is fast, parallel, and nearly effortless; route finding *between* regions is slow, serial, and memory-bound.

This is consistent with the maze-tracing character of within-floor planning (a perceptual task at which humans are quick) and with chunking theory: the corridor layout is processed as a spatial gestalt, not as discrete graph nodes [6]. It immediately explains the empirical shape of indoor difficulty guidelines, which grade courses almost entirely by counts of floor changes, stairs, and one-way passages — never by corridor complexity alone [1].

The consequence is that **the human's effective search graph is the cluster graph**: one node per floor region, one edge per labeled staircase or one-way passage. A venue with thousands of map positions collapses, for the planner, to a few dozen clusters and 10–20 stair edges. Human route finding is search on this small graph under severe resource limits.

3.2 Resource limits

On the cluster graph the human solver is bounded in three ways:

- **Working memory.** A planned route must be carried as a stair-label sequence (e.g. *dcccCA*: down D, down C three floors, up C, up A). At roughly four chunks [5], sequences beyond 4–6 crossings cannot be held reliably while running unless they are re-chunked (repeated letters compress well: *ccc* is one chunk) or re-read en route.

- **Exploration budget.** Each candidate stair considered costs a visual fixation on the map plus a cross-segment search for where the stair lands. Under race pressure a runner evaluates only a handful of candidates per decision, nothing like the systematic expansion of a best-first search.
- **No global comparison.** Comparing two complete multi-floor routes requires holding both — typically beyond capacity. Hence route *choice* degenerates to route *finding*: the first plan that closes is taken [7].

3.3 The observed strategy

From these constraints, the strategy reported by practitioners and consistent with the wayfinding literature follows almost by necessity:

1. **Same-floor first.** Check whether the leg can be solved within the current cluster. If yes, run it immediately — even when a stair route would be shorter. This is the floor strategy of Hölscher et al. [2] taken to its extreme, and it is cheap because within-cluster solving is perceptual.
2. **Bidirectional planning.** Otherwise, alternate between two easy local questions: *which stairs can I reach from here?* (forward) and *which stairs would drop me at the control?* (backward). Both are single-segment perceptual queries. Planning from both ends halves the depth that must be bridged — the classic advantage of bidirectional search, discovered here not by algorithm design but by memory pressure.
3. **Local heuristics for ordering.** Among candidate stairs, prefer the near one, and the one whose far end lies toward the goal (the direction strategy [2]).
4. **First connection wins.** When the forward and backward fragments meet in a common floor region, the plan closes: concatenate the forward labels with the backward labels in reverse discovery order, and start running. No alternatives are examined.

The result is satisficing — settling for a plan that is good enough rather than searching for the best one [7] — deterministic-ish, and fast — and systematically non-optimal in ways the algorithmic analysis can measure (Section 6).

3.4 Memory of previous routes

The search just described does not start from scratch on every leg. Each solved leg leaves behind reusable knowledge: which stair labels exist and roughly where they sit, which regions a given staircase connects, and which passages are one-way. A staircase already traversed is a known quantity — its far end no longer needs to be searched for on the other segment, so evaluating it as a candidate costs almost nothing. In chunking terms [6], earlier routes convert parts of the cluster graph from raw map content into pre-formed chunks, simultaneously widening the effective exploration budget and cheapening each planned crossing. This predicts two observable effects: later legs revisiting familiar stairs are planned faster than structurally equivalent early legs, and runners should show a bias toward stairs they have already used even when a slightly better unfamiliar alternative exists — a bias course setters can exploit in the same way as the commitment traps of Section 4.

4 Challenges

4.1 One-way passages

One-way passages break the symmetry that bidirectional planning relies on: a passage usable

in the forward fragment may be illegal when mentally walked backward from the control, and a plan that ignores this discovers the error only on the ground, where recovery is expensive (the passage cannot be re-entered against its direction). The technique that addresses this is *directional discipline in the backward pass*: when asking "which stairs/passages would bring me here?", only arrows pointing *into* the region count. This is exactly the point where novices' backward planning fails, and course setters exploit it deliberately; the difficulty guidelines gate one-way passages accordingly [1].

4.2 Discontinuous map segments

A staircase's two ends appear on different map segments, linked only by its label. Every crossing added to a plan therefore costs a visual search: find the label on the other segment, reorient, and continue. Errors here — matching the wrong label, or the right label on the wrong segment — corrupt the whole plan. Practitioners mitigate this with *label vocabulary* (memorizing the venue's stair letters and their approximate positions during the run-in or from earlier legs) and with map handling: thumb on the current position, map folded to keep the two active segments adjacent.

The problem sharpens when the segments span multiple *pages* — for large venues the map may be printed on both sides of the sheet, or on several sheets. A route crossing a page boundary then involves segments that cannot be viewed simultaneously at all: the runner must flip or swap the map mid-plan, losing both the thumb anchor on the current position and any visual context from the segment just left. Bidirectional planning suffers particularly, since the forward and backward fragments may live on different pages, forcing the working-memory-bound part of the plan to bridge a gap that no fold can close. Crossings between pages are therefore markedly more expensive than crossings between co-visible segments, and both course setters and difficulty models should treat the page boundary as an additional structural cost.

4.3 Working-memory overflow

Legs requiring many crossings exceed the 4-chunk budget. Copers use three techniques: **compression** (encode the plan as the letter string *dcccCA* and rehearse it verbally — repeated letters and familiar pairs compress into single chunks [6, 5]); **plan-as-you-go** (commit only to the first crossing or two, and re-plan at an intermediate region — trading map-reading time while moving for memory load); and **offloading to the map** (re-derive the remainder at each floor change rather than storing it). Which mix is optimal depends on running speed, map complexity, and the runner's rehearsal skill — a genuine individual-differences question the framework can test.

4.4 Commitment traps

Satisficing means the first closed plan wins, so a course setter can place an attractive early connection that is far from optimal — a *bait*. The countermeasure is a cheap *sanity bound*: before running, compare the plan's rough length or crossing count against the leg's straight-line distance; a plan that feels disproportionate warrants one more look. Experts appear to run this check habitually and almost instantly; it is a prime candidate for explicit training (Section 5).

4.5 Error recovery

When execution diverges from the plan (a missed one-way, a wrong stair), the runner must *relocate*: re-establish position, then re-plan from there. The same-floor-first strategy makes relocation cheaper than it looks — the current cluster is perceptually solvable once position is known — so the critical skill is recognizing *that*

one is off-plan early. Anticipatory map reading, the hallmark of expert orienteers [8], serves exactly this: knowing what should appear next makes deviations easy to notice.

5 Training the technique

The analysis above decomposes indoor route finding into trainable components. Deliberate practice — isolated components, immediate feedback, graded difficulty [9] — suggests the following program, ordered from perceptual to strategic. The open toolchain [10] supports several of these directly, and a companion training application [11] implements the program below as a set of deliberate-practice drills with timing and immediate feedback.

1. **Cluster reading.** Timed drills: given a map segment and two points, trace the within-floor route. Trains the perceptual solver that everything else builds on; speed and error rate are easy to measure.

2. **Stair vocabulary.** Given a segment, name all stairs and their directions (up/down) as fast as possible; then, from memory, point to a named stair. Builds the label–position mapping that cross-segment searches depend on.

3. **Label-sequence memory.** Rehearse routes as letter strings ($dcccCA$); recall them after a physical effort interval (memory under load differs sharply from memory at rest). Practice compression: find the chunking of a given string that survives a 200 m run.

4. **Backward planning.** From the control, list every stair/passage that can legally reach it, one-way arrows respected. Do this on paper courses until it becomes automatic.

5. **Bidirectional closure drills.** Full legs, planned under a countdown, answering only: forward reachable stairs, backward reaching stairs, first meeting region. The countdown forces satisficing; the drill trains making the *first* plan a good one by improving candidate ordering.

6. **Bait detection.** Review legs where the human-style plan and the optimal route diverge (the toolchain enumerates all routes and can compute both [10]); learn the structural patterns — attractive near stair, hidden one-way, shorter route via unintuitive direction — and practice the sanity bound.

7. **Post-race comparison.** After events, compare the route actually run (from split times or memory) against the enumerated route inventory: was the loss in planning (wrong route) or execution (right route, run badly)? This closes the deliberate-practice feedback loop at the level of whole races.

Course setters can invert the same list: each drill corresponds to a difficulty knob (cluster complexity, stair count and naming, sequence depth, one-way density, bait placement) that the graph model makes explicit and countable.

6 Simulating the human solver

The strategy of Section 3.3 is precise enough to implement. The analysis toolchain [10] contains a deterministic *human route* algorithm operating on the same venue graph as the optimal analyzers:

1. **Same-floor short-circuit:** if the leg is solvable through unlabeled corridors alone (one-way corridors included in their legal direction), return that route immediately.

2. **Alternating bidirectional search** on the cluster graph: one expansion per side per

turn, forward side first. Candidate stairs from the current region are ordered nearest-first, with straight-line direction to the goal as tie-break; each side visits a region at most once (a human does not re-explore).

3. **First valid meeting wins:** when a region is reached by both sides, forward labels are concatenated with the backward side's labels in reverse discovery order; a meeting whose two fragments would reuse the same physical staircase is rejected and the search continues. The concrete path is stitched through cached within-region shortest paths, and the search stops.

The simulation is intentionally non-optimal, produces exactly one route, and is deterministic, so its predictions are testable. It enables measures the optimal analysis cannot provide: the *human–optimal gap* (length ratio between the simulated human route and the true shortest route — a direct bait detector); *plan depth* (crossings in the plan, a working-memory load estimate); and *mental steps* (expansions before the plan closed, a planning-time estimate). A leg can be easy for the simulation and hard for Dijkstra-based scoring, or vice versa; systematic divergence between the two, compared against split-time data, is a sharp empirical test of whether human difficulty is better predicted by the human model. A companion plan in the repository describes the intended refinements: exposing the exploration trace for narration, parameterizing the cognitive limits (memory span, exploration budget, heuristic weights), and fitting those parameters per runner class against event data [10].

7 Research directions

1. **Validate the same-floor bias:** from split times, measure how often runners take within-floor routes when a shorter stair route exists, by class and by time pressure.

2. **Fit cognitive parameters:** treat memory span, exploration budget, and candidate-ordering weights as free parameters of the simulation; fit them to observed route choices per runner class.

3. **Human-model difficulty scoring:** use plan depth, mental steps, and the human–optimal gap as terms in a difficulty model, and compare its predictive power against structural models on the same data [1].

4. **Bait taxonomy:** classify legs with a large human–optimal gap by their structural cause (near-stair bait, one-way trap, direction illusion); this gives course setters a designed-difficulty vocabulary.

5. **Training studies:** test whether the drills of Section 5 transfer — e.g. does label-sequence memory training reduce time loss specifically on deep-sequence legs?

6. **Instrumented stairwells:** place additional split-time units inside stairwells, not just at course controls, to measure time spent per floor change directly rather than inferring it, and to separate planning delay (deciding which stair to take) from execution delay (running it) within a single leg.

8 Conclusion

Human indoor-orienteering route finding is bounded search on the condensed cluster graph: floor regions are solved at a glance and treated as single chunks, while stair-sequence planning is deliberative, bidirectional, satisficing (Section 3.3), and limited by a working memory of a few chunks. The hard parts of indoor navigation — one-way passages, discontinuous

map segments, deep stair sequences, commitment traps — are exactly the points where this strategy's assumptions break, and each has an identifiable countermeasure and a corresponding training drill. Because the strategy is precise enough to simulate, human-style route finding can now be computed alongside optimal routes, turning claims about "what a runner would do" into measurable, falsifiable quantities.

References

[1] *A graph-based framework for quantitative analysis of navigational complexity in indoor orienteering courses*. Companion article, same repository.

[2] Hölscher, C., Meilinger, T., Vrachliotis, G., Brösamle, M., & Knauff, M. (2006). Up the down staircase: Wayfinding strategies in multi-level buildings. *Journal of Environmental Psychology*, 26(4), 284–299.

[3] Wiener, J. M., Büchner, S. J., & Hölscher, C. (2009). Taxonomy of human wayfinding tasks: A knowledge-based approach. *Spatial Cognition & Computation*, 9(2), 152–165.

[4] Miller, G. A. (1956). The magical number seven, plus or minus two: Some limits on our capacity for processing information. *Psychological Review*, 63(2), 81–97.

[5] Cowan, N. (2001). The magical number 4 in short-term memory: A reconsideration of mental storage capacity. *Behavioral and Brain Sciences*, 24(1), 87–114.

[6] Chase, W. G., & Simon, H. A. (1973). Perception in chess. *Cognitive Psychology*, 4(1), 55–81.

[7] Simon, H. A. (1956). Rational choice and the structure of the environment. *Psychological Review*, 63(2), 129–138.

[8] Eccles, D. W., Walsh, S. E., & Ingledew, D. K. (2002). A grounded theory of expert cognition in orienteering. *Journal of Sport and Exercise Psychology*, 24(1), 68–88.

[9] Ericsson, K. A., Krampe, R. T., & Tesch-Römer, C. (1993). The role of deliberate practice in the acquisition of expert performance. *Psychological Review*, 100(3), 363–406.

[10] *Indoor Orienteering Route Analysis* (source code, algorithm reference, and datasets). `github:objorke/indoor-o-route-analysis`.

[11] *Indoor-O Training* (companion deliberate-practice training application). `github:objorke/indoor-o-training`.